THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** lustre	Metals reflect light.
2	**(b)** brittle	They break when hammered.
3	**(c)** metal oxide	E.g. magnesium oxide.
4	**(b)** malleability	E.g. aluminium foil.
5	**(d)** oxygen	Oxygen is a gas.
6	**(b)** basic	They turn red litmus blue in water.
7	**(b)** ductility	E.g. copper wire.
8	**(b)** bromine	Bromine is a reddish-brown liquid.
9	**(d)** acidic	E.g. sulphur dioxide.
10	**(c)** sonority	E.g. bells.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Lustre and malleability (1); Ductility and conductivity (1).	1+1
12	Mercury (1); Bromine (1).	1+1
13	It conducts electricity (1); It is soft and marks paper (1).	1+1
14	Malleable: can be beaten into sheets (1); Ductile: can be drawn into wires (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Make a circuit with a gap (1); Place metal and non-metal samples (1); Metals light the bulb; non-metals do not (except graphite) (1).	1+1+1
16	Malleable (1); Good heat conductor (1); Does not corrode easily (1).	1+1+1
17	Metal + acid gives salt + hydrogen (1); E.g. $\mathrm{Mg + 2\,HCl \longrightarrow MgCl_2 + H_2}$ (1); Hydrogen burns with a pop (1).	1+1+1
	OR: Lustre: metals shiny, non-metals dull (1); Conductors vs insulators (1); Malleable vs brittle (1).	1+1+1
18	Very reactive (1); Reacts with air and moisture (1); Kerosene protects it (1).	1+1+1
	OR: A base is formed (1); $\mathrm{MgO + H_2O \longrightarrow Mg(OH)_2}$ (1); Turns red litmus blue (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Lustre: gold (1); Malleability: aluminium (1); Ductility: copper (1); Conductivity: silver (1); Sonority: bell metal (1).	1+1+1+1+1
	OR: Mercury liquid (1); Sodium soft (1); Graphite conducts (1); Iodine lustrous (1); Diamond hard (1).	1+1+1+1+1
20	Non-metals brittle (1); Dull (1); Poor conductors (1); Not sonorous (1); Metals are opposite (1).	1+1+1+1+1
	OR: Copper: conductor (1); Aluminium: light, malleable (1); Iron: strong (1); Gold: lustrous, unreactive (1); Use matches property (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that all metals are hard and solid; **mercury** is liquid and **sodium** is soft.
- Saying that all non-metals are poor conductors; **graphite** conducts electricity.
- Mixing up oxides: metal oxides are *basic* (red litmus to blue), non-metal oxides are *acidic* (blue litmus to red).
- Confusing malleability (sheets) with ductility (wires).
- Forgetting that rusting needs **both** air and moisture, and that galvanisation uses *zinc*.
- Writing that a less reactive metal can displace a more reactive one from its solution (e.g. copper displacing iron).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** iodine	Iodine has a shiny appearance.
2	**(b)** magnesium oxide	$2\,Mg + O_2 \longrightarrow 2\,MgO$.
3	**(d)** copper	Metals conduct.
4	**(d)** high melting points	Most metals melt at high temperatures.
5	**(a)** sulphur dioxide	$S + O_2 \longrightarrow SO_2$.
6	**(c)** sulphur	Sulphur is a brittle yellow solid.
7	**(d)** poor conductors of heat	Sulphur, wood-like carbon.
8	**(b)** red	It is acidic.
9	**(b)** mercury	It is used in thermometers.
10	**(a)** a good conductor and ductile	Used widely.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Burns with a dazzling white flame (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1).	1+1
12	They are brittle (1); They break when stretched (1).	1+1
13	Metals: iron, copper (1); Non-metals: sulphur, oxygen (1).	1+1
14	Metal oxides are basic (1); Non-metal oxides are acidic (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Mercury is a liquid metal (1); Sodium is soft (1); Graphite conducts (1).	1+1+1
16	Oxygen: breathing (1); Sulphur: gunpowder, matches (1); Carbon: fuel (1).	1+1+1
17	Non-metals do not displace hydrogen (1); They are less reactive in this way (1); No gas is formed (1).	1+1+1
	OR: Metals conduct (1); Ductile to make wires (1); Plastic insulates (1).	1+1+1
18	Metals conduct heat (1); Wood is a poor conductor (1); So handles are of wood or plastic (1).	1+1+1
	OR: CO2 is a non-metal oxide (1); It forms an acid (1); Turns blue litmus red (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); Oxides are basic (1); $Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Hydrogen is released (1); Hydrogen burns with a pop (1).	1+1+1+1+1
	OR: Allow current and heat (1); Tester circuit (1); Bulb glows (1); Heat a rod and watch wax (1); Pin falls (1).	1+1+1+1+1
20	$S + O_2 \longrightarrow SO_2$ (1); $C + O_2 \longrightarrow CO_2$ (1); Oxides are acidic (1); Dissolve in water (1); Turn blue litmus red (1).	1+1+1+1+1
	OR: Look for lustre (1); Hammer it (1); Test conduction (1); Test sound (1); Combine results (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that all metals are hard and solid; **mercury** is liquid and **sodium** is soft.
- Saying that all non-metals are poor conductors; **graphite** conducts electricity.
- Mixing up oxides: metal oxides are *basic* (red litmus to blue), non-metal oxides are *acidic* (blue litmus to red).
- Confusing malleability (sheets) with ductility (wires).
- Forgetting that rusting needs **both** air and moisture, and that galvanisation uses *zinc*.
- Writing that a less reactive metal can displace a more reactive one from its solution (e.g. copper displacing iron).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **The World of Metals and Non-metals** | Class 7 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** blue	It is basic.
2	**(b)** graphite	Graphite is a form of carbon.
3	**(c)** malleable	It can be rolled into thin sheets.
4	**(a)** hydrogen gas	E.g. zinc and HCl.
5	**(b)** diamond	Diamond is a form of carbon.
6	**(c)** good conductors of heat	Heat spreads quickly.
7	**(c)** zinc chloride and hydrogen	A salt and hydrogen gas.
8	**(a)** sodium	Sodium is soft.
9	**(a)** sonorous	They ring.
10	**(d)** pop sound	Hydrogen.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Foil for wrapping (1); Utensils or wires (1).	1+1
12	Metals conduct heat well (1); Food cooks quickly (1).	1+1
13	$Zn + 2\,HCl \longrightarrow ZnCl_2 + H_2$ (1); Hydrogen gas is evolved (1).	1+1
14	They are lustrous (1); They resist corrosion (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Magnesium oxide is basic (1); Sulphur dioxide is acidic (1); Litmus changes accordingly (1).	1+1+1
16	Metals produce sound when struck (1); E.g. bell (1); Non-metals do not ring (1).	1+1+1
17	Test lustre (1); Test conduction (1); Test hammering (1).	1+1+1
	OR: Reddish-brown copper deposits (1); Blue colour fades (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1).	1+1+1
18	Strong and hard (1); Can be shaped (1); Last long (1).	1+1+1
	OR: Utensils (1); Wires (1); Vehicles (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Take copper sulphate solution (1); Place an iron nail (1); Brown deposit forms (1); $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$ (1); Colour changes to green (1).	1+1+1+1+1
	OR: Copper age (1); Bronze (1); Iron age (1); Machines (1); Modern alloys (1).	1+1+1+1+1
20	Iron: bridges (1); Copper: wires (1); Aluminium: foil (1); Oxygen: breathing (1); Carbon: fuel (1).	1+1+1+1+1
	OR: $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1); Basic (1); $C + O_2 \longrightarrow CO_2$ (1); Acidic (1); Test using litmus (1).	1+1+1+1+1

Mistakes to avoid:

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- Mixing up oxides: metal oxides are *basic* (red litmus to blue), non-metal oxides are *acidic* (blue litmus to red).
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